The PV logger comm fail: what the controller does today, and the small change that stops it

Erbil plant, InteliNeo 5500 master, PLC sheet 10. Written from the archive and history exported on 14 October 2026.

What happens today

Every tenth of a second the controller decides how much it wants from the solar plant and writes it to the SAU. Sheet 10 is the last stage: it keeps the request within 100 kW of what the inverters already give, ramps it, sends it. It also holds a small switch, Item 76. Normally the request passes through; but the moment either PV logger session misses a single reply, the switch flips to its other position, which is a fixed zero. The inverters obey within seconds, the gensets jump from 25 percent to the whole plant, the UPS breakers are shed, and the solar then needs about five minutes to climb back, 100 kW at a time. Five to thirty times a day in the history. And the zero never reaches the logger while the link is down; it arrives when the link is back, when there is nothing left to protect.

The change

We leave the switch, the ramp and every protection. The switch's second input is wired to its own output instead of the fixed zero, so when a reply is missed it simply repeats the last value it sent. The inverters see no drop, the gensets see no step. The block in front of it, Item 68, no longer fires on a request of zero (that guard would now keep an old value when the controller really wants zero), only on the two comm fail signals. And one new timer raises a flag, PV comm lost, after five minutes of silence, for an alarm now and, if the plant agrees later, for opening the PV breakers from sheet 3.

Is holding safe?

While the link is down the inverters keep their last setpoint in both programs; the old zero never had a say. The gensets keep 25 percent of their nominal in hand (788 kW with three running) against a load drop, the controller's own protections do not depend on the logger, and after five minutes the flag is up.

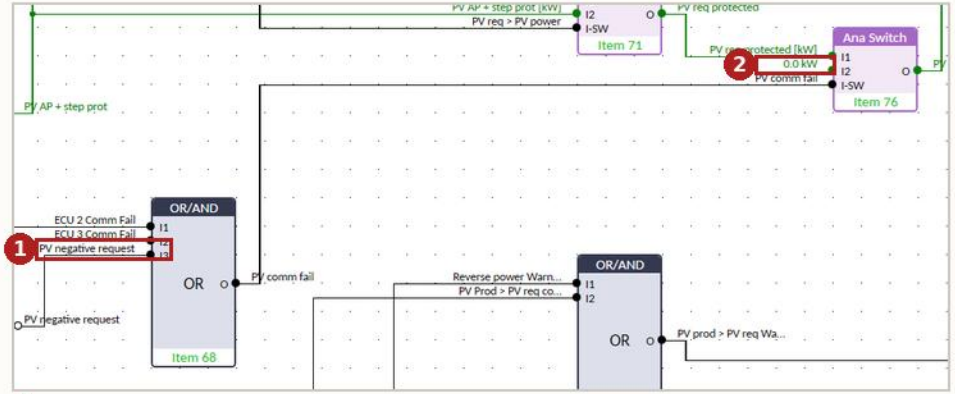
Checked

The whole program was run block by block against the plant's history. With a healthy link the new sheet gives the same outputs as the old one, to the kilowatt. On 14 October it removes three of the eleven UPS sheddings and gives back about 350 kWh of solar; the rest was the OFF/AUTO toggles.

Loading, testing, the way back

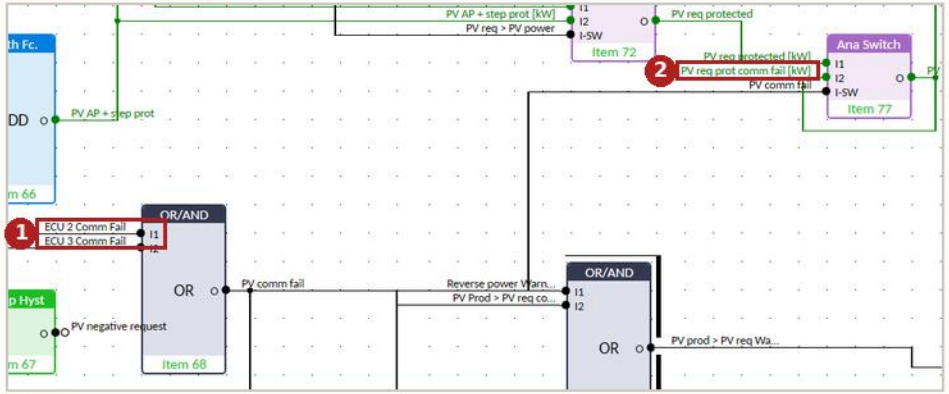
Copy of the archive, import sheet10_PV_req_step_protection_v2_const.xml, check the three marked places, compare, save as a new archive. Test at low load with the plant's agreement: SAU cable out 30 s, the solar must not move; out 6 min, the flag comes up, the UPS stay. Undo: the 14 October archive, or the constant 0.0 kW back on the switch.

Today



1. Item 68 fires on a missed reply, and also on a request of zero or below.
2. Item 76: the second position is a fixed 0.0 kW. That is what goes out.

After the change



1. Item 68 reacts only to ECU 2 and ECU 3 comm fail.
2. The switch (now Item 77) has its own output on its second input: it holds the last value.



3. New: Item 65, a 300 s timer on PV comm fail, raises the flag PV comm lost (sheet output 5).